

Multi-variable Linear Regression

Lab (7)

Lab 1: Boston Housing Pricing

- Context: Predict median house values using neighborhood features.
- Dataset: `fetch_openml(name='boston', version=1)`
- Tasks:
 1. Load dataset and split into 80% train / 20% test
 2. Scale features using `StandardScaler`
 3. Train linear regression model
 4. Calculate MSE and R^2 on test set
 5. Print top 3 most important features

Lab 2: Diabetes Progression Prediction

- Context: Predict disease progression using multiple health indicators.
- Dataset: `load_diabetes()` (scikit-learn)
- Tasks:
 1. Load dataset and examine feature distributions
 2. Create feature interaction: BMI $\times$ Blood Pressure
 3. Train model with/without interaction term
 4. Compare R^2 scores for both models
 5. Plot residuals vs predictions

Lab 3: California Housing Prices

- Context: Predict median house values using California census data.
- Dataset: `fetch_california_housing()` (scikit-learn)
- Tasks:
 1. Add new feature: Rooms per person (`AveRooms / Population`)
 2. Handle outliers in `MedInc` (cap at 97th percentile)
 3. Train model and calculate VIF for multicollinearity
 4. Plot feature importance
 5. Predict price for: `MedInc=4.0`, `HouseAge=20`, `AveRooms=5`, `Population=1200`, `AveOccup=2.5`

Lab 4: Wine Quality Prediction

- Dataset: [Red Wine Quality](#) (winequality-red.csv)
- Tasks:
 1. Create new features:
 - Acid balance: citric acid / volatile acidity
 - Sulfur ratio: free SO₂ / total SO₂
 2. Check correlation matrix and remove one feature from pairs with $|r| > 0.8$
 3. Train model and evaluate with 10-fold cross-validation
 4. Identify most important chemical property

Lab 5: Car Price Prediction

- Dataset: [Car Features and MSRP](#) (cars.csv)
- Tasks:
 1. Preprocess:
 - Log-transform MSRP
 - One-hot encode categorical features
 2. Add feature: Fuel efficiency = $0.5 \times (\text{Highway MPG} + \text{City MPG})$
 3. Train model and interpret coefficients
 4. Calculate price premium for sports cars vs SUVs

Lab 6: House Price Prediction

- Dataset: [House Prices - Advanced Regression](#) (train.csv)
- Tasks:
 1. Handle missing values (mean for numeric, mode for categorical)
 2. Create age feature: $2023 - \text{YearBuilt}$
 3. Train Ridge regression model ($\alpha=1.0$)
 4. Compare performance with OLS linear regression
 5. Identify most valuable neighborhood

Lab 7: Bike Rental Demand

- Dataset: [Bike Sharing Hourly](#) (bike_sharing_hourly.csv)
- Tasks:
 1. Cyclically encode hour:

```
X['hour_sin'] = np.sin(2 * np.pi * X['hour']/24)
```

```
X['hour_cos'] = np.cos(2 * np.pi * X['hour']/24)
```

2. One-hot encode weather and season
3. Train model and analyze peak demand hours
4. Calculate demand drop during heavy rain

Lab 8: Concrete Strength Prediction

- Dataset: [Concrete Strength](#) (concrete.csv)
- Tasks:
 1. Create feature: Water/Cement ratio
 2. Check multicollinearity using Variance Inflation Factor (VIF)
 3. Train model with polynomial features (degree=2)
 4. Compare R^2 with simple linear model

Lab 9: Student Writing Score Prediction

- Dataset: [Student Performance](#) (students.csv)
- Tasks:
 1. Create feature: TotalScore = MathScore + ReadingScore
 2. One-hot encode categorical features
 3. Train model and interpret gender coefficient
 4. Calculate expected writing score for:
 - o Math=85, Reading=90, TestPrep="completed"

Lab 10: Medical Insurance Cost Prediction

- Dataset: [Medical Insurance Costs](#) (insurance.csv)
- Tasks:
 1. Create interaction: Smoker $\times$ BMI
 2. One-hot encode region
 3. Train model and compare with/without interaction
 4. Calculate cost difference:
 - o Smoker vs non-smoker with BMI=30
 - o 60-year-old vs 30-year-old